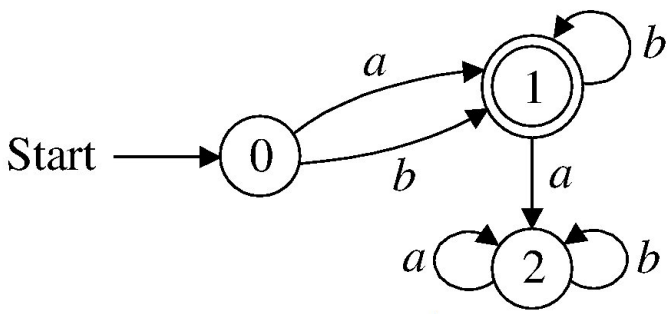
Finite Automata

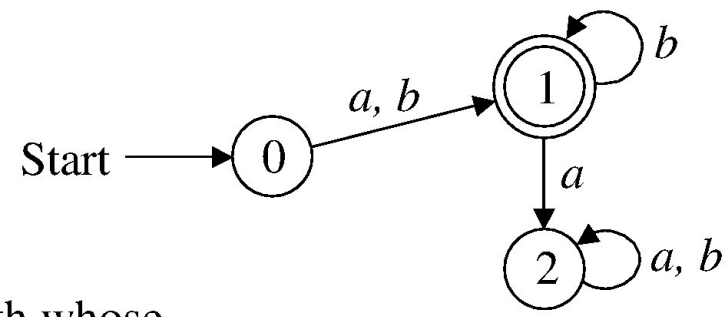
For academic learning

Deterministic Finite Automata

A *deterministic finite automaton* (DFA) over an alphabet A is a finite digraph (where vertices or nodes are called *states*) for which each state emits one labeled edge for each letter of A. One state is designated as the *start state* and a set of states may be *final states*.

Example. Either of the following alternatives is acceptable for representing a DFA, where final states are indicated by double circles.





The *Execution* of DFA for input string $w \in A^*$ begins at the start state and follows a path whose edges concatenate to w. The DFA *accepts* w if the path ends in a final state. Otherwise the DFA *rejects* w. The *language* of a DFA is the set of accepted strings.

Example. The example DFA accepts the strings

$a, b, ab, bb, abb, bbb, \ldots, ab^n, bb^n, \ldots.$

Nondeterministic Finite Automata

A *nondeterministic finite automaton* (NFA) over an alphabet A is similar to a DFA except that Λ-edges are allowed, there is no requirement to emit edges from a state, and multiple edges with the same letter can be emitted from a state.